

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 y_u^{11i2} \text{LF}_{2,2,0} [m_2, \tilde{\mu}] - \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_u^{11i2} \text{LF}_{3,2,-1} [m_2, \tilde{\mu}] + \right. \\ & \frac{1}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} \text{LF}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{p}}] + \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{p}}] + \\ & \frac{1}{2} \tilde{\mu} g_2^2 y_e^{\text{pr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] - \frac{1}{4} \tilde{\mu} g_1^2 y_e^{\text{pr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\text{p}}, m_e^{\text{r}}] - \\ & \frac{1}{8} \tilde{\mu} y_e^{\text{pr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} (3 g_1^2 + 7 g_2^2) \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{6} \tilde{\mu} y_e^{\text{pr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} (3 g_1^2 + 2 g_2^2) \\ & \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{2} \tilde{\mu} y_d^{\text{pr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} (2 g_1^2 + 3 g_2^2) \text{LF}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] - \\ & \frac{1}{4} \tilde{\mu} g_1^2 y_d^{\text{pr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_d^{\text{r}}] - \frac{21}{8} \tilde{\mu} y_d^{\text{pr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + \\ & \frac{1}{2} \tilde{\mu} y_d^{\text{pr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] + \\ & \frac{1}{8} \tilde{\mu} y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} (g_1^2 - g_2^2) \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] - \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\text{p}}, m_u^{\text{r}}] + \\ & \frac{1}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_u^{\text{r}}] + \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_u^{\text{r}}] - \\ & \frac{1}{4} \tilde{\mu} y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} (g_1^2 - 2 g_2^2) \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] + \\ & \frac{1}{8} \tilde{\mu} y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} (-g_1^2 + g_2^2) \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{p}}] - \\ & \frac{3}{4} \tilde{\mu} y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} (g_1^2 + 2 g_2^2) \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] + \frac{1}{2} \tilde{\mu} y_u^{\text{pr}} y_u^{11i2} \overline{\text{au}}^{\text{pr}} \\ & (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (g_1^2 + g_2^2) \text{LF}_{3,1,0} [\tilde{\mu}, m_1] - \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_u^{11i2} \text{LF}_{3,2,-1} [\tilde{\mu}, m_1] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (6 g_1^2 + 5 g_2^2) \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_u^{11i2} \text{LF}_{4,2,-2} [\tilde{\mu}, m_1] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \\ & \frac{1}{12} m_2 \tilde{\mu} g_2^2 y_u^{11i2} (9 g_1^2 + 5 g_2^2) \text{LF}_{3,1,0} [\tilde{\mu}, m_2] - \frac{7}{3} m_2 \tilde{\mu} g_2^4 y_u^{11i2} \text{LF}_{3,2,-1} [\tilde{\mu}, m_2] - \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^2 y_u^{11i2} (18 g_1^2 + 11 g_2^2) \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + \\ & 2 m_2 \tilde{\mu} g_2^4 y_u^{11i2} \text{LF}_{4,2,-2} [\tilde{\mu}, m_2] + \frac{1}{2} m_2 \tilde{\mu} g_2^2 y_u^{11i2} (3 g_1^2 + 2 g_2^2) \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1} [m_1, m_q^{i1}, m_u^{i2}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1} [m_1, m_u^{i2}, m_q^{i1}] + \\ & \frac{1}{48} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (-7 g_1^2 + g_2^2) \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] + \\ & \frac{1}{12} m_1 \tilde{\mu} g_1^2 y_u^{11i2} (7 g_1^2 - g_2^2) \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] - \\ & \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 y_u^{11i2} (7 m_1 + 5 m_2) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_1] + \\ & \tilde{\mu} g_1^2 g_2^2 y_u^{11i2} (m_1 + m_2) \text{LF}_{3,2,1,-2} [m_2, \tilde{\mu}, m_1] - \frac{3}{16} m_2 \tilde{\mu} g_2^2 y_u^{11i2} (g_1^2 - 7 g_2^2) \\ & \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] - \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_u^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1} [m_3, m_q^{i1}, m_u^{i2}] - \\ & \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_u^{11i2} (g_1^2 + g_2^2) \text{LF}_{2,2,1,-1} [m_3, m_u^{i2}, m_q^{i1}] + \frac{3}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} \\ & \text{LF}_{2,2,1,-1} [m_d^{\text{r}}, m_q^{\text{p}}, m_d^{\text{t}}] + \frac{1}{24} \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} \overline{\text{ad}}^{\text{pr}} (7 g_1^2 + 3 g_2^2) \text{LF}_{2,1,1,0} [m_d^{\text{r}}, m_q^{\text{p}}, \tilde{\mu}] - \\ & \frac{1}{8} \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} \overline{\text{ad}}^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{3,1,1,-1} [m_d^{\text{r}}, m_q^{\text{p}}, \tilde{\mu}] - \frac{1}{16} \tilde{\mu} y_d^{11r} y_u^{\text{pi}2} \overline{\text{ad}}^{\text{pr}} (g_1^2 + g_2^2) \\ & \text{LF}_{2,2,1,-1} [m_d^{\text{r}}, \tilde{\mu}, m_q^{\text{p}}] - \frac{3}{4} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} y_u^{11i2} \overline{\text{ad}}^{\text{pr}} \text{LF}_{2,2,1,-1} [m_d^{\text{t}}, m_q^{\text{p}}, m_d^{\text{r}}] + \\ & \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{2,2,1,-1} [m_e^{\text{r}}, m_l^{\text{p}}, m_e^{\text{t}}] - \frac{1}{4} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \\ & \text{LF}_{2,2,1,-1} [m_e^{\text{t}}, m_l^{\text{p}}, m_e^{\text{r}}] - \frac{1}{2} \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{2,1,1,0} [m_l^{\text{p}}, m_e^{\text{r}}, m_e^{\text{t}}] + \\ & \text{LF}_{2,2,1,-1} [m_l^{\text{p}}, m_l^{\text{s}}, m_e^{\text{r}}] - \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{2,1,1,0} [m_l^{\text{s}}, m_e^{\text{r}}, m_l^{\text{p}}] + \\ & \tilde{\mu} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} y_u^{11i2} \overline{\text{ae}}^{\text{pr}} \text{LF}_{3,1,1,-1} [m_l^{\text{s}}, m_e^{\text{r}}, m_l^{\text{p}}] - \frac{3}{2} \tilde{\mu} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} y$$